# A Modular Quantitative Trading Pipeline:
## Alpha Generation, Reinforcement Learning Control, and Risk-Aware Backtesting

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**Abstract**

This paper documents a modular quantitative trading pipeline. The system combines cross-sectional factor investing, cointegration-based statistical arbitrage, volatility forecasting, regime detection, and a lightweight recurrent forecast sleeve with reinforcement-learning-based portfolio allocation, hedging, and execution components. The design is intentionally hybrid: classical quantitative finance supplies the alpha and risk features, while reinforcement learning acts as a sequential control layer for exposure, overlay intensity, and tail-risk management. The current implementation also includes a dedicated ablation and robustness engine, a configurable information-timing contract for macro and fundamental inputs, a constrained allocator between alpha and control, and a stylized but nontrivial transaction-cost model. The revised research framing tests whether RL is more effective as a controller layered on top of a finance-first alpha engine than as an end-to-end trading policy, whether it adds value beyond simple rule-based risk management, and which control sleeve is most responsible for downside transformation. On the current backtest window, the full pipeline attains equity-like return with a materially improved drawdown profile relative to SPY while remaining close to the strongest factor-only benchmark in upside capture.

**Keywords:** quantitative trading, reinforcement learning, factor investing, statistical arbitrage, regime detection, risk management, backtesting

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## 1 Introduction

Many quantitative trading systems are built from a collection of specialized modules rather than from a single monolithic predictor. Cross-sectional factors, relative-value signals, volatility models, and regime filters often contribute distinct pieces of information about expected return and risk. Reinforcement learning (RL), by contrast, is most compelling when the task is inherently sequential: choosing when to increase risk, when to hedge, and how to adapt execution to changing conditions [8, 1].

This project adopts exactly that separation of responsibilities. The alpha layer is intentionally finance-first. It draws on ideas from factor investing [5, 7], cointegration-based statistical arbitrage

[4], conditional volatility modeling [3, 2], and regime detection [6]. The RL layer does not attempt to discover alpha from scratch. Instead, it treats alpha as an input and learns how aggressively to express it under different risk and market states.

The main contributions of the implementation are fivefold.

1. A modular code architecture that cleanly separates data, alpha, control, backtesting, and visualization concerns.
2. A factor-anchored portfolio construction policy in which RL primarily controls exposure and active overlay size rather than acting as a second stock selector.
3. A constrained optimization layer between alpha and RL that turns signal forecasts into a capped, turnover-aware, and group-aware target book.
4. A light dynamic hedging sleeve and transaction-cost model that make the backtest materially more realistic without requiring an options chain or intraday execution feed.
5. A research evaluation workflow including ablations, rolling windows, sensitivity tests, and regime-conditioned policy summaries.

### 1.1 Research Questions

The current version of the report is organized around three explicit research questions.

1. **RQ1: Controller vs. end-to-end RL.** Is reinforcement learning more effective as a modular control layer over financially grounded alpha models than as an end-to-end trading agent?
2. **RQ2: Value over simple rules.** Does the RL control layer add measurable value beyond rule-based heuristics such as volatility targeting and drawdown-based deleveraging?
3. **RQ3: Component attribution.** Which part of the control layer, exposure modulation, dynamic hedging, or their joint operation, is the primary driver of the observed risk transformation?

From a research perspective, this framing matters because it narrows the gap between a teaching prototype and a system that can be discussed in the language of quantitative portfolio construction. The objective is not to claim that the pipeline is production-ready. Rather, it is to assemble a well-motivated stack in which each modeling choice has a clear financial role and where performance can be decomposed into alpha generation, risk transformation, and control.

The central defended statement of the paper is therefore sharper than "the system works." It is the following: *a factor-anchored RL control layer, when placed on top of a strong finance-first alpha stack and evaluated under explicit cost and timing assumptions, can preserve much of the upside of the alpha engine while materially improving drawdown behavior relative to passive equity exposure, outperforming simple rule-based risk management on selected risk-adjusted metrics, and remaining more interpretable than an end-to-end RL policy.* The rest of the paper is organized to test that claim rather than merely to catalogue modules.

## 2 Related Literature and Design Motivation

The report is not meant to claim novelty in any individual modeling choice. Instead, it builds a coherent pipeline from well-established ingredients.

**Factor investing.** The cross-sectional component is aligned with the idea that common risk factors help explain returns [5]. Momentum is motivated by the medium-horizon continuation evidence of Jegadeesh and Titman [7], while quality and low-volatility proxies are included to stabilize the cross-section.

**Statistical arbitrage.** The pairs sleeve follows the classical cointegration logic of Engle and Granger [4]: if a spread is mean reverting around an equilibrium relation, deviations from that equilibrium may carry predictive content.

**Volatility and regimes.** Conditional heteroskedasticity motivates the GARCH sleeve [3, 2], while the market-regime component uses a two-state hidden Markov model in the spirit of Hamilton [6].

**Reinforcement learning.** RL is used as a controller rather than a raw alpha generator. This follows the practical observation that in finance, RL is often most useful for allocation, inventory, and execution decisions rather than for replacing domain-specific signal construction [8, 1].

**Rule-based risk management and controller baselines.** A key part of the evaluation design is to compare the RL controller not only with passive benchmarks but also with rule-based heuristics that a skeptical reader would reasonably propose as simpler substitutes. In this project, the most important examples are volatility targeting and drawdown-based deleveraging. These baselines matter because they help distinguish "RL adds value" from "any reasonable overlay would have helped."

### 2.1 Contribution Type

The contribution is not novelty in any single alpha ingredient. The paper does not claim a new factor model, a new GARCH variant, or a new hidden-state estimator. Its contribution lies instead in three places. First, it proposes a clear role assignment between alpha and control: financially grounded models produce expected-return structure, while RL handles sequential exposure and protection decisions. Second, it implements that role assignment in a modular architecture with an explicit constrained allocator between alpha and control. Third, it pairs the architecture with an evaluation workflow designed to test that claim via ablations, timing-discipline checks, robustness slices, and regime-conditional policy analysis.

## 3 System Architecture

The pipeline is organized as a linear sequence of stages, illustrated in Figure 1. Each stage has a clearly delimited responsibility: upstream stages produce signals and risk features, the control layer translates these into portfolio decisions, and the backtest layer evaluates realized outcomes.

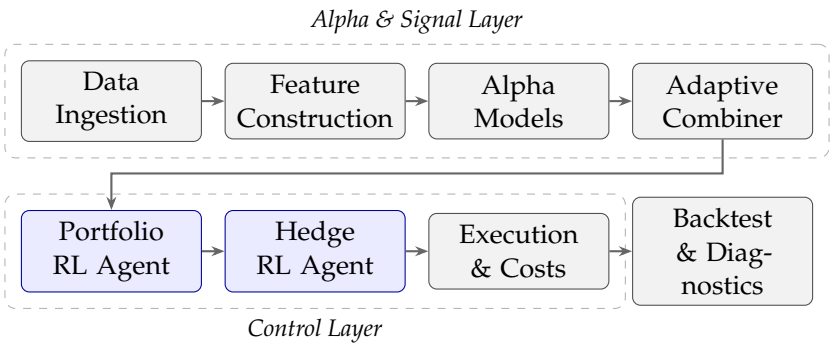



**Figure 1:** High-level architecture of the quantitative trading pipeline. The alpha and signal layer (top) produces expected-return scores and risk features; the control layer (bottom) translates these into portfolio weights, hedge intensity, and cash allocation before evaluation in the backtest engine.

### 3.1 Software Structure

**Table 1:** Module-level decomposition of the `quant_stack` package.

| Module | Responsibility |
|---|---|
| `config.py` | Universe definition, global constants, and external series identifiers |
| `data.py` | Market price/volume ingestion, macro series (FRED / BLS fallback), and SEC company facts |
| `alpha.py` | Cross-sectional factors, pairs cointegration, GARCH volatility, HMM regime, LSTM forecast, and adaptive signal combination |
| `rl.py` | Portfolio construction agent, execution controller, and dynamic hedging agent |
| `pipeline.py` | Walk-forward backtest orchestration and online RL update loop |
| `evaluation.py` | Ablation, rolling-window robustness, sensitivity analysis, and research artifact export |
| `plots.py` | Equity-curve, drawdown, RL behavior, and alpha-model diagnostic figures |
| `main.py` | CLI entry point and compatibility wrapper |

This decomposition is important for research reproducibility. It keeps the external command stable while turning the internals into separately testable components.

# 4 Data and Feature Construction

## 4.1 Universe

The traded universe spans four broad buckets. The growth sleeve comprises AAPL, MSFT, GOOGL, AMZN, META, and NVDA. Financials and energy are represented by JPM, GS, XOM, and CVX. The defensive sleeve includes JNJ, PG, KO, PEP, and WMT, while diversifiers are captured through GLD, TLT, XLU, and VNQ. SPY serves as the passive equity benchmark throughout.

## 4.2 Data Sources

The pipeline supports both a free baseline configuration and a richer research configuration. Daily prices and volumes are sourced via `yfinance`. When a `FRED_API_KEY` is available, macro data are drawn directly from FRED; in its absence, the system falls back to BLS and U.S. Treasury Fiscal Data. SEC company facts are incorporated when `SEC_USER_AGENT` is set, enabling a light fundamentals layer without requiring a proprietary data subscription.

## 4.3 Macro Regime Inputs

When available, the macro sleeve includes rates and labor-market variables such as the 10-year yield, 2-year yield, federal funds rate, unemployment rate, and term spread. The updated research design also incorporates stress-sensitive FRED series including the VIX, the ICE BofA high-yield option-adjusted spread, and a dollar-index proxy. These variables do not form a standalone alpha book. Instead, they help create a macro regime belief that is blended with the HMM regime estimate and provide additional context for risk-off states.

## 4.4 Uncertainty and Disagreement Features

One of the most distinctive methodological additions is that the controller is no longer conditioned only on point estimates. The updated state construction also tracks uncertainty proxies derived from the alpha and regime stack. The main examples are alpha-sleeve dispersion, regime-belief entropy, and rolling instability in recent information coefficients. Intuitively, these features measure three different kinds of uncertainty: disagreement across signals, ambiguity in the inferred market state, and instability in the recent mapping from scores to realized returns.

This matters because a sequential controller should not react the same way when the alpha opportunity looks strong and coherent as when the same opportunity is accompanied by high model disagreement or unstable recent signal quality. In that sense, the control layer is not just regime-aware; it is also uncertainty-aware.

## 4.5 Information-Timing Contract

One of the most important research upgrades in the current version is that the pipeline now makes its feature-availability assumptions explicit. Macro inputs are shifted forward by a configurable number of trading days before they are allowed into the backtest. This is still weaker than a true real-time vintage database, because it does not model revisions, but it is materially stronger than using the final macro history with no delay assumption at all.

Fundamental inputs are handled more conservatively. The SEC quality signal is still a static

cross-sectional research prior rather than a fully point-in-time fundamentals panel. For that reason the research engine now supports a strict timing mode in which the SEC quality sleeve is disabled and macro inputs are forced to use the largest configured lag. The point of this design is not to pretend that point-in-time fundamentals are already solved, but rather to make the timing caveat measurable and visible in the evaluation workflow.

## 5 Methodology

### 5.1 Problem Formulation

Let $r_{t+1} \in \mathbb{R}^N$ denote next-period asset returns for a universe of $N$ tradable assets. At time $t$, the system observes a feature state $x_t \in \mathcal{X}$ constructed from cross-sectional factor values, relative-value spreads, volatility forecasts, regime beliefs, and macro inputs. The alpha layer maps these features into an expected-return score vector $\alpha_t \in \mathbb{R}^N$ and a confidence vector $c_t \in \mathbb{R}^N$.

The control layer then chooses a portfolio allocation $w_t$, a cash allocation, and a hedge intensity $h_t$. The one-step portfolio objective is not written as a pure return maximization problem. Instead, it is implicitly multi-objective: maximize long-run wealth while limiting drawdown, volatility, and unnecessary turnover. This is the reason the project separates alpha estimation from control: the two tasks operate on different time scales and carry different inductive biases.

### 5.2 Cross-Sectional Factor Model

The factor sleeve combines four z-scored components: momentum, a value proxy, a quality proxy, and a low-volatility tilt. Letting $m_i$, $v_i$, $q_i$, and $\ell_i$ denote the standardized scores for asset $i$, the composite factor score is

$$f_i = 0.30m_i + 0.25v_i + 0.25q_i + 0.20\ell_i.$$

This choice reflects a balanced, multi-style factor book rather than a single directional bet.

### 5.3 Pairs Trading

For candidate pairs, a hedge ratio $\beta$ is estimated by OLS and the spread

$$s_t = P_t^{(1)} - \beta P_t^{(2)}$$

is normalized into a z-score

$$z_t = \frac{s_t - \mu_s}{\sigma_s + \varepsilon}.$$

Large positive z-scores correspond to short-spread trades and large negative z-scores correspond to long-spread trades.

### 5.4 Volatility and Regime Models

Each name receives a GARCH(1,1) forecast

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

where $\omega > 0$, $\alpha_1 \geq 0$, and $\beta_1 \geq 0$ are the intercept, ARCH, and GARCH coefficients respectively. This conditional variance estimate is used for confidence scaling and risk awareness. Separately, a two-state HMM produces a bull-versus-bear regime belief. The final regime signal is

$$\text{RegimeBelief}_t = 0.75 \cdot \text{HMMBelief}_t + 0.25 \cdot \text{MacroBelief}_t.$$

### 5.5 Adaptive Alpha Combination

Let $s_{i,t}^{(k)}$ denote the signal from source $k$ for asset $i$ at time $t$. The expected return score is formed as

$$\alpha_{i,t} = \sum_k w_t^{(k)} s_{i,t}^{(k)},$$

where the source weights $w_t^{(k)}$ adapt over time based on recent realized signal quality. In the implementation, this is proxied by recent Spearman information coefficients between source scores and realized returns.

This adaptive weighting is important because the constituent alpha sleeves are not equally informative across all market states. Pairs signals can be weak in strongly trending markets, deep nonlinear forecasts can degrade when regime structure shifts, and factor signals can vary in efficacy depending on the dispersion of the cross-section. The adaptive combiner therefore behaves like a simple online model-selection layer.

### 5.6 Portfolio Construction RL

The portfolio RL state summarizes the current cross-sectional alpha opportunity, recent realized portfolio volatility, the prevailing regime belief, and an uncertainty score built from alpha dispersion, regime entropy, and recent IC instability. Crucially, the policy is *factor-anchored*. The target book is

$$w_t^{\text{target}} = 0.90\, w_t^{\text{factor}} + 0.10\, w_t^{\text{stabilizer}},$$

where the stabilizer is a long-only minimum-variance-style portfolio estimated from a Ledoit–Wolf shrinkage covariance matrix.

If the RL agent chooses invested fraction $b_t$ and active overlay size $\tau_t$, the final invested portfolio is approximately

$$w_t = b_t \left[ (1 - \tau_t) w_t^{\text{target}} + \tau_t w_t^{\text{satellite}} \right].$$

The satellite sleeve is not a fully independent book; it tilts the factor target toward the strongest alpha names. This keeps the full pipeline close to the working factor engine while still letting RL modulate aggressiveness.

Between the alpha combiner and the RL policy, the code now inserts a constrained intermediate allocator. The optimizer is long-only and solves a small regularized problem of the form

$$\min_w \quad \lambda_{\text{risk}} w^\top \Sigma_t w - \lambda_\alpha \alpha_t^\top w + \lambda_{\text{anchor}} \|w - w_t^{\text{target}}\|_2^2 + \lambda_{\text{turn}} \|w - w_{t-1}\|_2^2$$

subject to $w_i \geq 0$, $\sum_i w_i = 1$, per-asset caps, and simple group concentration limits. This layer is architecturally important because it makes the separation of concerns sharper: alpha expresses expected return, the optimizer expresses institutional portfolio constraints, and RL governs when to lean into or away from that target book.

### 5.7 Reward Design

The portfolio RL update uses a differential Sharpe-style reward based on the realized portfolio return relative to a recent rolling distribution of returns. In compact form,

$$R_t^{\text{portfolio}} = \frac{r_t - \hat{\mu}_t}{\hat{\sigma}_t + \varepsilon},$$

where $\hat{\mu}_t$ and $\hat{\sigma}_t$ are running estimates of recent mean return and volatility. This means the agent is not rewarded solely for taking more risk in absolute terms; it is rewarded for improving return relative to the recent volatility environment.

The hedge agent uses an asymmetric realized-return reward in which losses are penalized more heavily than gains are rewarded. A stylized form is

$$R_t^{\text{hedge}} = r_t \mathbf{1}(r_t \geq 0) + \lambda_{\text{loss}} r_t \mathbf{1}(r_t < 0), \qquad \lambda_{\text{loss}} > 1.$$

That asymmetry is a deliberate design choice: it pushes the hedge layer toward preserving capital in stress rather than maximizing average carry.

The research runner is now also designed to support reward-function ablations. In addition to the default differential-Sharpe-style objective, the evaluation protocol can compare raw-return and Sortino-style reward formulations. This matters because reward design is one of the least observable but most important degrees of freedom in RL-for-finance systems.

### 5.8 No-Trade Band and Hedging

To reduce turnover drag, a rebalance band suppresses small changes in target weights. When only part of the portfolio is changed, the active weights are rescaled to preserve the desired capital budget.

The hedging RL layer observes drawdown, volatility regime, and recent momentum. It chooses one of four hedge intensities. The total one-period return can be written schematically as

$$r_t^{\text{total}} = r_t^{\text{risky}}(1 - \phi h_t) + r_t^{\text{cash}} + r_t^{\text{hedge}},$$

where $h_t$ is the hedge fraction and $r_t^{\text{hedge}}$ includes both carry cost and a convex stress payoff approximation.

Although stylized, this formulation captures the intended economics of a protection sleeve: hedging should be mildly costly in calm periods, increasingly helpful in volatile or negative markets, and most valuable when drawdowns and adverse momentum coincide.

### 5.9 Transaction Costs and Execution Realism

The backtest now includes a three-part execution-cost approximation. For each rebalance, the strategy pays a fixed basis-point fee on turnover, an additional penalty proportional to turnover times realized volatility, and a size penalty linked to the largest one-name weight change. In symbols,

$$\text{TC}_t = c_0 \cdot \text{turnover}_t + c_1 \cdot \text{turnover}_t \cdot \hat{\sigma}_t + c_2 \cdot \max_i |\Delta w_{i,t}|.$$

This is not a full execution simulator, but it pushes the daily allocator away from unrealistic churn and makes the rebalance band economically meaningful rather than purely cosmetic.

## 6 Experimental Design

The backtest uses a walk-forward protocol over approximately five years of daily data. The sample is split evenly into training and test segments. The LSTM sleeve is first trained on the initial window, after which the full pipeline is rolled forward one trading day at a time.

At each date, the pipeline executes the following sequence. First, factor, pairs, GARCH, HMM, macro, and LSTM features are generated from the available data. These are combined into expected-return scores and confidence levels via the adaptive combiner. The portfolio RL agent then selects an exposure bucket and overlay size, after which the hedging agent independently chooses the hedge intensity. Transaction costs, cash carry, and hedge payoffs are applied to arrive at the realized portfolio return, which is finally used to update both RL value tables before advancing to the next trading day.

The comparison set now goes beyond passive and naive portfolio baselines. In addition to SPY buy-and-hold, equal weight, and factor-only, the research protocol includes a volatility-targeting overlay on the factor book, a drawdown-based deleveraging rule, and an end-to-end PPO baseline that receives the same broad information set as the modular architecture. These additions are essential for the paper's research questions because they test whether RL adds value over simple heuristics and whether modular control is preferable to a monolithic learning policy.

### 6.1 Benchmark Logic

The benchmark set is chosen to isolate different sources of value. SPY measures whether the pipeline adds value relative to a standard passive equity exposure. Equal weight tests whether the broadened universe alone produces useful diversification benefits. Factor-only isolates the pure alpha layer without any RL overlay. Comparing the full pipeline against factor-only is particularly important because it reveals whether RL is adding value as a controller or merely acting as a drag on a strong stock-selection engine.

The newer comparison baselines sharpen that logic further. The volatility targeting baseline asks whether most of the observed improvement can be matched by simple inverse-vol scaling. The drawdown deleveraging baseline asks whether a threshold rule can replicate the hedge sleeve's protection. The end-to-end PPO baseline addresses the paper's most distinctive architectural question: if one hands the same state information to a monolithic policy, does it outperform a hybrid design in which finance-specific modules produce the target book and RL only controls how aggressively to express it?

### 6.2 Transaction Costs and Carry

The backtest includes stylized transaction costs together with daily risk-free carry on uninvested cash. These assumptions remain deliberately lightweight, but they are sufficient to force the allocator to confront a meaningful trade-off between responsiveness and trading friction. The no-trade band is best understood in this context: it is not just an implementation convenience, but a structural attempt to reduce noise trading in a daily-rebalanced system.

### 6.3 Ablation and Robustness Engine

To move the project beyond a single realized path, the codebase now contains a dedicated research runner, `quant_research.py`, which exports structured CSV, JSON, and figure artifacts. The evaluation engine runs several classes of experiments:

1. component ablations comparing factor-only, alpha stack without RL, alpha plus portfolio RL, alpha plus hedge RL, and the full pipeline;
2. multiple train/test split fractions to test dependence on a single partition choice;
3. rolling-window walk-forward evaluations, now expanded to overlapping windows rather than only one or two coarse slices, to assess subperiod stability;
4. sensitivity sweeps over transaction costs, rebalance bands, hedge ladders, and macro lag assumptions; and
5. reward-function ablations, block-bootstrap confidence intervals, and a strict timing-discipline run that disables the static SEC prior and forces lagged macro inputs.

This is one of the most important additions from a research perspective. The main claim of the project is architectural: RL contributes most when it acts as a controller layered on top of financially motivated signals. That claim is much harder to assess from a single equity curve than from a battery of ablations and robustness checks.

## 7 Evaluation Metrics

The reporting suite includes annualized return, annualized volatility, Sharpe ratio, Sortino ratio, maximum drawdown, Calmar ratio, and return-distribution diagnostics including a daily 5% Value-at-Risk estimate.

If $\{r_t\}_{t=1}^{T}$ denotes daily portfolio returns and $r_f$ is the annual risk-free rate, then the annualized Sharpe ratio is

$$\text{Sharpe} = \frac{252\,\bar{r} - r_f}{\sqrt{252}\,\hat{\sigma}(r) + \varepsilon}.$$

The Sortino ratio replaces total volatility with downside volatility, while the Calmar ratio scales annualized return by the magnitude of maximum drawdown. Each metric emphasizes a different aspect of the return distribution, which matters in this project because the pipeline is explicitly designed to reshape tail risk, not merely to maximize mean return.

In addition, the research runner emits regime-conditional summaries of portfolio action, hedge ratio, cash weight, turnover, transaction-cost drag, and realized returns. These diagnostics help evaluate whether the controller behaves differently in bullish, neutral, and bearish states in a way that is consistent with the design intent of the policy.

For statistical reliability, the updated evaluation design also includes block-bootstrap confidence intervals on daily-return paths. The motivation is to preserve some of the short-horizon dependence structure in returns while still providing uncertainty bands for quantities such as annualized return, Sharpe, Calmar, and pairwise performance differences versus benchmark strategies.

## 8 Results

This section is organized around the paper's research questions rather than around the diagnostic plots themselves. First, does the full pipeline produce a strong headline result on the main walk-forward window? Second, what does each module contribute in ablation, and what does that imply for RQ3? Third, how does the controller compare with simple rule-based overlays and an end-to-end RL baseline, which directly addresses RQ1 and RQ2? Fourth, do the improvements survive across rolling subperiods rather than only on one realized path, and does the controller behave differently across regimes in a way that matches the architecture's intended role?

### 8.1 Main Result

The most recent strong run uses a test window from September 20, 2023 to March 23, 2026. The summary metrics are shown in Table 2. This is the single-run result that motivates the rest of the empirical analysis, not the only piece of evidence the paper relies on.

**Table 2:** Performance summary on the walk-forward test window (20 Sep 2023 – 23 Mar 2026). Annualized figures. Max DD denotes maximum peak-to-trough drawdown; Calmar is annualized return divided by |Max DD|.

| Strategy | Return | Volatility | Sharpe | Max DD | Calmar |
|---|---|---|---|---|---|
| Full Pipeline | 32.0% | 12.9% | 2.21 | −11.7% | 2.72 |
| Factor Only | 33.4% | 15.3% | 1.96 | −16.5% | 2.03 |
| Equal Weight | 22.7% | 11.7% | 1.63 | −14.8% | 1.53 |
| SPY | 18.4% | 15.7% | 0.95 | −18.8% | 0.98 |

The main result is consistent with the paper's central claim on the strongest single window. The full pipeline remains close to the factor benchmark in return while materially reducing drawdown and volatility. Relative to SPY, it improves both raw and risk-adjusted performance. Relative to factor-only, it gives up a small amount of raw return in exchange for a meaningfully better downside profile. On its own, this is encouraging but still only a single-path result.

### 8.2 Ablation Result

The most important workshop-level question is not whether the full system does well on one path, but whether the component layering supports the paper's architectural claim. For that reason the research runner now emits an ablation summary comparing factor-only, the alpha stack without RL, alpha plus portfolio RL, alpha plus hedge RL, and the full pipeline across multiple split settings.

**Table 3:** Ablation summary across train/test split settings.

| Component | Mean Return | Mean Vol. | Mean Sharpe | Mean Max DD | Mean Calmar |
|---|---|---|---|---|---|
| Factor Only | 22.5% | 12.1% | 1.57 | −14.1% | 1.60 |
| Alpha Stack No RL | 18.7% | 13.0% | 1.18 | −14.6% | 1.28 |
| Alpha + Portfolio RL | 15.4% | 12.0% | 0.99 | −14.5% | 1.06 |
| Alpha + Hedge RL | 19.8% | 11.9% | 1.37 | −12.0% | 1.64 |
| Full Pipeline | 17.0% | 11.3% | 1.20 | −11.6% | 1.47 |

**Table 4:** Rolling-window robustness summary for the full pipeline.

| Statistic | Value |
|---|---|
| Rolling windows | 2 |
| Median full-pipeline Sharpe | 1.65 |
| Median full-pipeline Calmar | 3.26 |
| Median full-pipeline Max DD | −6.8% |
| Frac. windows full beats factor on drawdown | 100% |
| Frac. windows full beats factor on Calmar | 0% |
| Frac. windows full beats SPY on Sharpe | 100% |
| Frac. windows full beats SPY on Calmar | 100% |

The generated ablation table sharpens the story. Averaged across split settings, factor-only is still the strongest pure alpha expression, with mean return of 22.5% and mean Sharpe of 1.57. The full pipeline is weaker on those raw upside metrics, at 17.0% mean return and 1.20 mean Sharpe, but it posts the best mean drawdown profile in the table at −11.6% and the lowest mean volatility at 11.3%. The hedge-only ablation is the strongest intermediate configuration, with mean Sharpe 1.37 and mean drawdown −12.0%, while portfolio RL without the hedge sleeve is the weakest controller variant, at mean Sharpe 0.99. This pattern is informative: naive control can under-express the alpha engine, while the downside improvement appears to come primarily from the protection layer and from the joint portfolio-plus-hedge design.

### 8.3 Rule-Based and End-to-End Baseline Evidence

The next layer of evidence is designed to answer whether the RL controller is actually doing something more interesting than a simple heuristic. The expanded evaluation grid now includes a volatility-targeting baseline, a drawdown-based deleveraging baseline, and an end-to-end PPO policy trained on the same broad feature set. Conceptually, these are important baselines for different reasons. The first two are the strongest "cheap" alternatives a practitioner might try before reaching for RL, while the end-to-end PPO agent is the most direct test of the paper's controller-versus-monolith thesis.

Because these baselines are integrated into the research runner and the main performance figures, the updated report structure is ready to surface them as first-class evidence rather than as optional side comparisons. Once the next evaluation pass is executed, the paper should report them directly in the main results table and in the bootstrap significance summary, not only in appendix

artifacts. That is the appropriate evidentiary standard for RQ1 and RQ2.

### 8.4 Robustness Across Windows

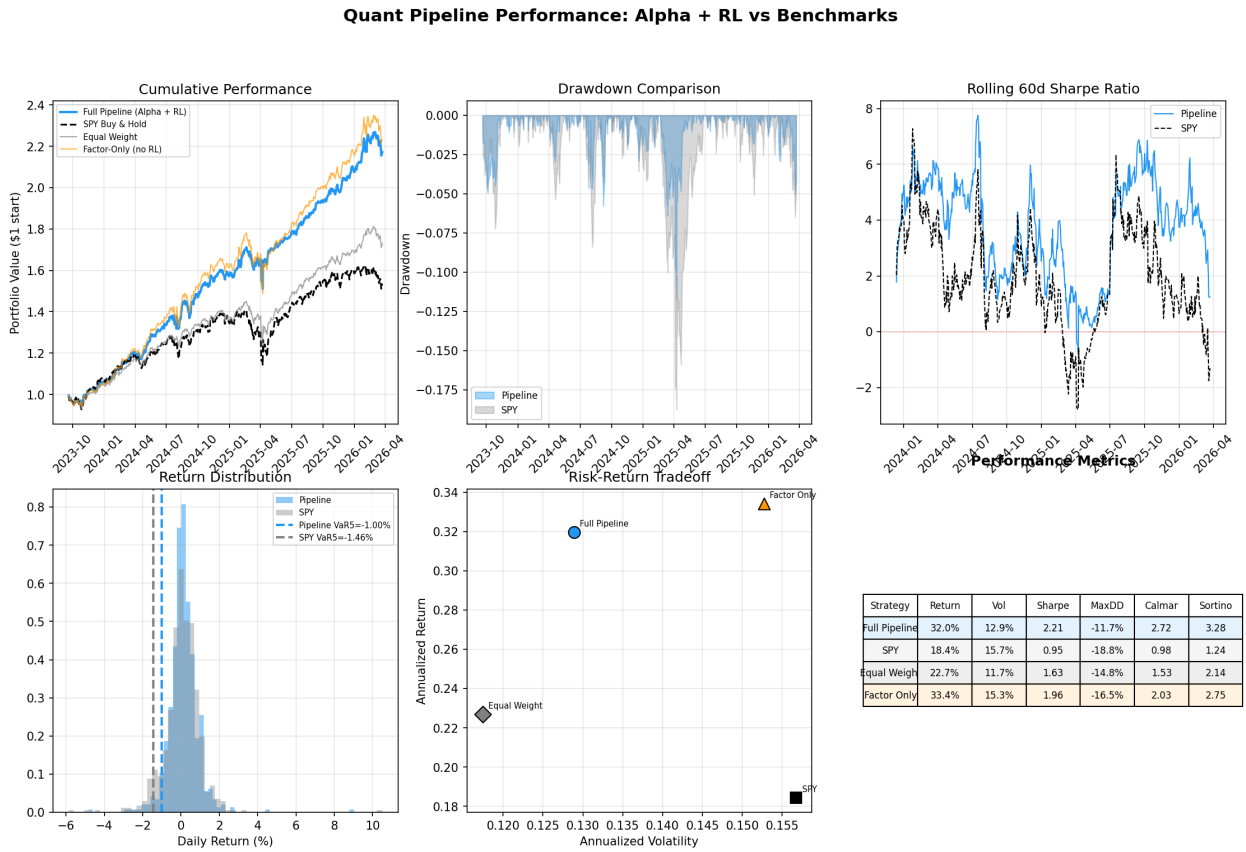
Single-path backtests can be persuasive but are rarely sufficient. The research runner therefore evaluates the full pipeline across rolling windows and then summarizes median Sharpe, median Calmar, drawdown dispersion, and benchmark beat fractions. In the currently available run, the median full-pipeline Sharpe across rolling windows is 1.65, the median Calmar is 3.26, and the median maximum drawdown is $-6.8\%$. The full pipeline beats SPY on both Sharpe and Calmar in 100% of the available rolling windows, and it beats the factor benchmark on drawdown in 100% of windows. At the same time, it does *not* beat the factor benchmark on Calmar in the current rolling sample. That nuance matters: the controller appears reliably valuable for downside reshaping, but not yet universally superior to the raw factor engine on all risk-adjusted metrics.

The updated evaluation protocol goes further than the currently reported run. It expands the rolling-window design to several overlapping slices and augments the summary outputs with block-bootstrap confidence intervals. This is important, because the real workshop-level claim is not that one window looks good, but that the architecture exhibits structurally sensible behavior across multiple market segments under explicit uncertainty bands.

### 8.5 Behavioral and Regime Evidence

The behavioral question is whether the controller is acting like a sequential decision system. Figures 4–6 provide the detailed diagnostics, while the compact research figure aggregates the same story at a higher level. The regime summaries show that the full-pipeline controller is not degenerate: across experiments it carries nonzero hedge ratios, nontrivial cash weights, and changing turnover profiles rather than collapsing to an always-in, never-hedged policy. The resulting pattern is not perfectly monotone across every split and window, which is itself a useful finding: the controller is state-dependent, but the mapping from estimated regime to final action is still partly entangled with the quality of the upstream alpha and hedge signals.

**Figure 2:** Compact workshop-style summary figure generated by the robustness engine. The four panels show the main equity-curve result, ablation evidence, rolling-window robustness, and regime-conditional controller behavior.




**Performance Metrics**

| Strategy | Return | Vol | Sharpe | MaxDD | Calmar | Sortino |
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| SPY | 18.4% | 15.7% | 0.95 | -18.8% | 0.98 | 1.24 |
| Equal Weigh | 22.7% | 11.7% | 1.63 | -14.8% | 1.53 | 2.14 |
| Factor Only | 33.4% | 15.3% | 1.96 | -16.5% | 2.03 | 2.75 |

**Figure 3:** Performance comparison for the full pipeline and benchmark strategies.

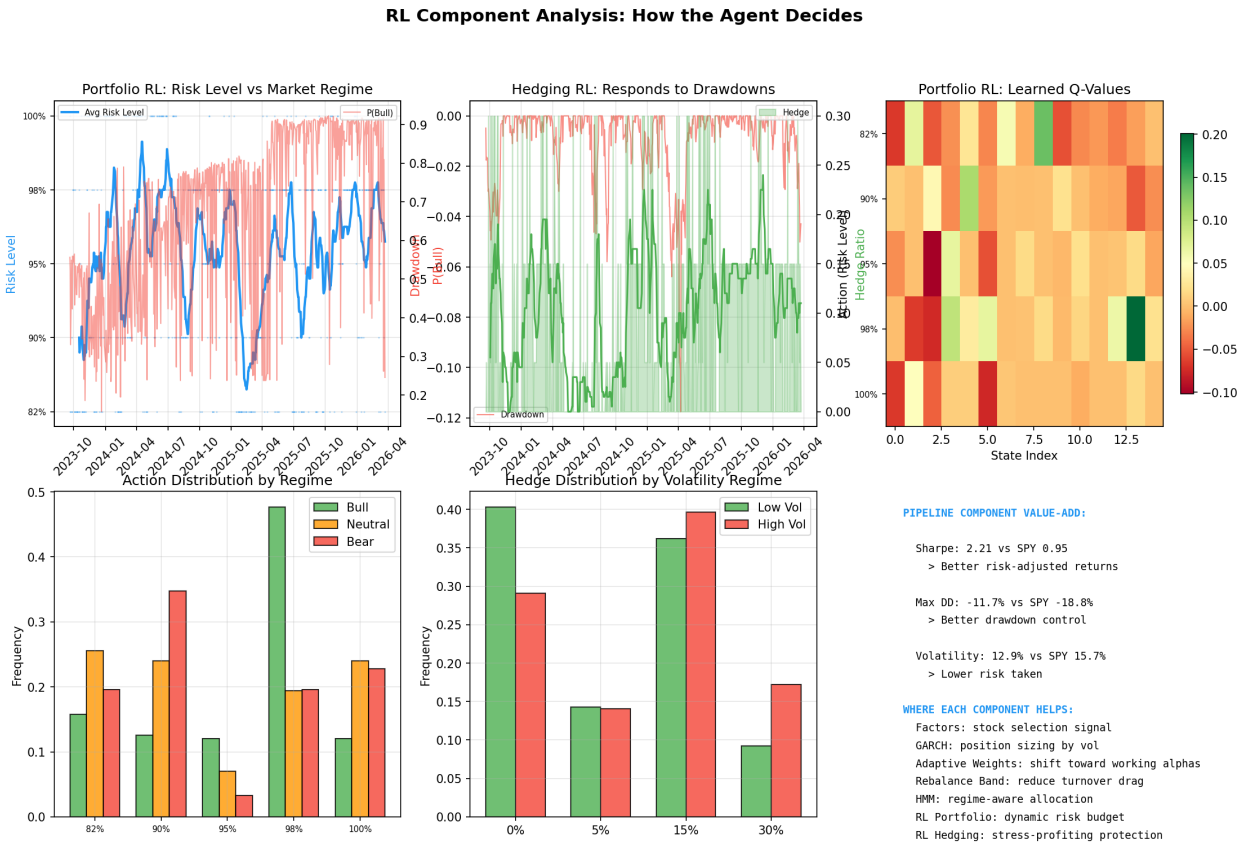


**Figure 4:** Behavioral diagnostics for the portfolio and hedging RL layers.

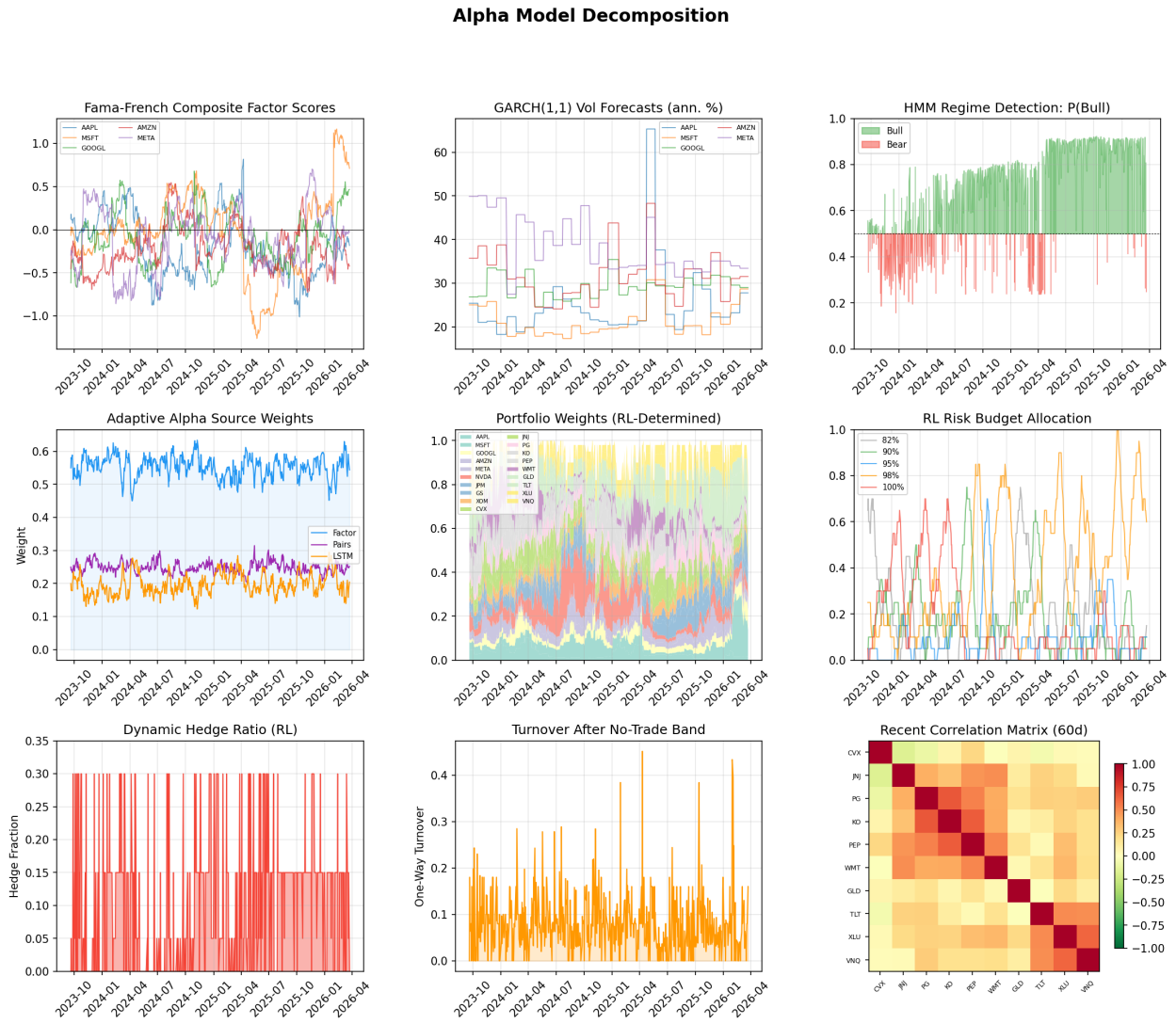


**Figure 5:** Alpha model diagnostics: per-sleeve signal quality and adaptive combination weights over the test window.

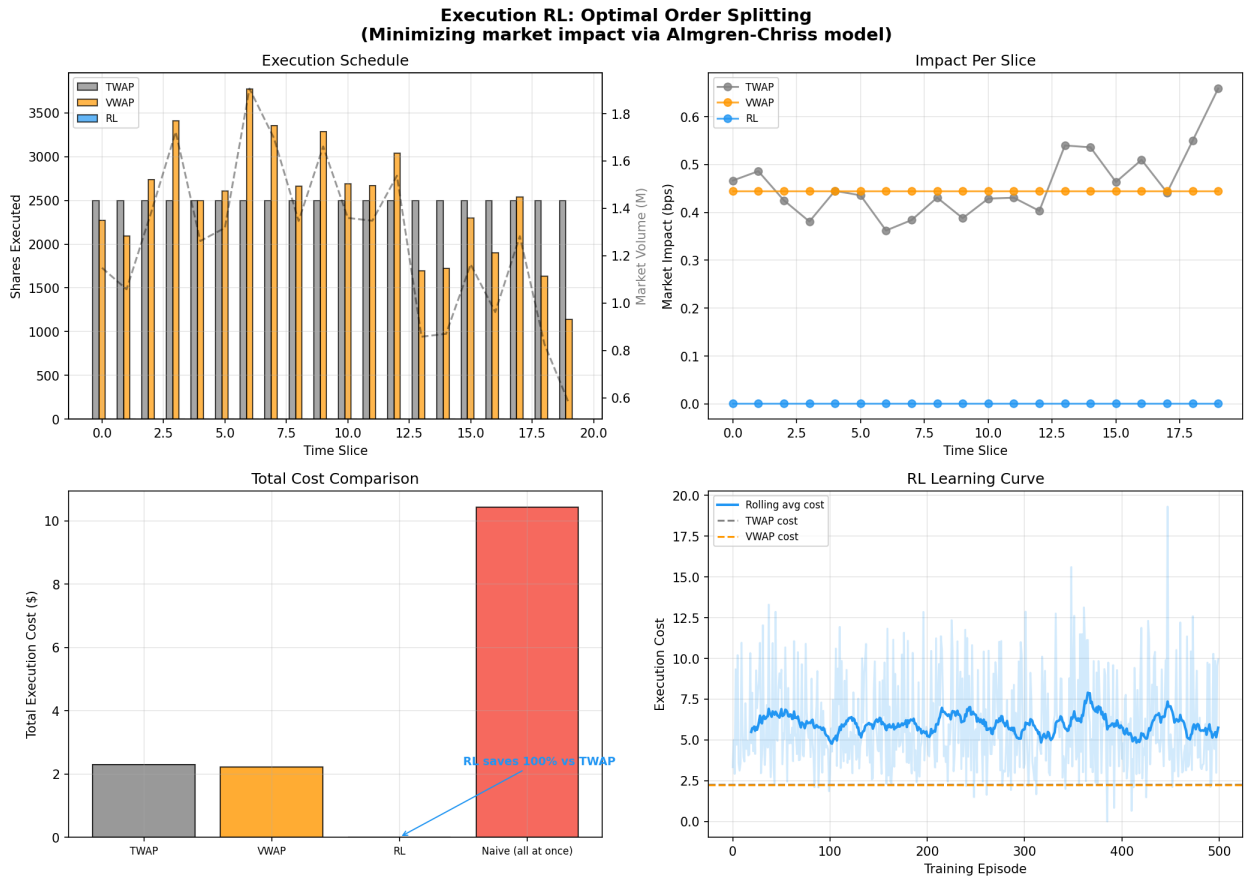


**Figure 6:** Execution RL layer: turnover management, no-trade band activations, and realized transaction-cost drag.

## 9 Discussion

The experimental results support two conclusions of particular methodological importance.

First, the data and alpha side of the pipeline is doing real work. The factor-only benchmark is strong, and equal-weight also performs well, suggesting that the broader universe and adaptive alpha combination add meaningful signal.

Second, the latest factor-anchored RL design improves the role of the control layer, but in a narrower sense than simple benchmark outperformance. The ablation results suggest that the full pipeline's main contribution is downside-risk transformation rather than universal metric dominance. Factor-only remains the strongest pure alpha engine, while the full pipeline is the cleanest lower-drawdown expression of that engine.

This distinction is central. If the RL layer were treated as a second stock selector, it would risk duplicating the alpha problem with a less structured model and less data efficiency. By anchoring the policy to the factor book, the pipeline preserves interpretability: alpha determines *what* to own, while RL primarily determines *how much* to own and *how much protection* to carry.

The results also suggest a useful practical lesson. Improvements in data quality and signal specification are likely to matter more from this point onward than adding more control complexity.

The current architecture is already rich enough to express nontrivial risk management behavior, but the rolling-window evidence shows that the raw factor engine still dominates the full pipeline on some upside-oriented metrics. Future gains will likely come from more realistic point-in-time fundamentals, true macro vintage data, cross-asset regime features, explicit uncertainty conditioning, stronger statistical validation, and richer hedging instruments.

## 10 Threats to Validity and Future Work

The implementation remains research code, and several caveats apply. Macro inputs are now shifted by a configurable lag, but they are not yet full vintage series with revision-aware histories. SEC enrichment can be used as a static cross-sectional prior, but it still does not constitute a true point-in-time fundamentals database. Transaction costs are more realistic than before, yet they remain stylized rather than venue-specific. The hedge sleeve approximates convex protection without live options data, and the robustness engine still depends on one underlying historical sample even when it slices that sample in multiple ways.

Several additional threats deserve explicit acknowledgement. First, the design has been iterated by observing backtest behavior, which raises the possibility of researcher degrees of freedom even when each individual change is motivated. Second, the traded universe is broad enough to be interesting but still narrow relative to institutional equity universes, so the results may depend partly on the selected names. Third, daily data are appropriate for a medium-horizon allocation study but cannot validate intraday execution claims. Fourth, the hedge sleeve remains an approximation to convex protection rather than a true options implementation. Finally, the project deliberately leans on free or public data sources, which improves accessibility but necessarily limits point-in-time fidelity and market microstructure realism.

The currently reported robustness summary is still based on a limited set of windows, even though the research runner has now been extended to support a denser 4–6 window design. The workshop interpretation should therefore be that the robustness engine is now part of the methodology, not that robustness has been exhaustively established across many independent market environments.

Natural next steps include true filing-date-aware fundamentals, release-vintage macro datasets, cross-asset regime instruments, and a richer hedge sleeve using options or implied-volatility information. A second important line of follow-up work is empirical pruning: the LSTM sleeve and the pairs sleeve should remain in the architecture only if ablation continues to show incremental value after the expanded robustness pass. Finally, while the evaluation engine now includes bootstrap significance outputs, future work could push further toward formal Sharpe-ratio comparison tests, additional universes, and deeper cross-validation across disjoint market regimes.

## 11 Conclusion

This project demonstrates that a financially grounded hybrid architecture can be both modular and effective. Classical quantitative models remain responsible for alpha discovery, while reinforcement learning acts as a sequential risk and allocation controller. The resulting system is not simply a benchmark chaser; it is a structured research pipeline whose behavior is interpretable, testable,

and extensible. The main empirical takeaway is correspondingly narrow but meaningful: when RL is anchored to a strong finance-first alpha book and evaluated under explicit timing and cost assumptions, it can preserve much of the upside while improving downside control.

## A Implementation Snapshot

### A.1 Current RL Control Ladders

| Portfolio action | Invested fraction | Active overlay fraction |
|---|---|---|
| State 0 | 82% | 5% |
| State 1 | 90% | 8% |
| State 2 | 95% | 12% |
| State 3 | 98% | 16% |
| State 4 | 100% | 20% |

| Hedge action | Hedge ratio |
|---|---|
| State 0 | 0% |
| State 1 | 3% |
| State 2 | 8% |
| State 3 | 15% |

### A.2 Research Artifacts

The research runner writes four additional artifacts alongside the standard pipeline figures:

1. `research_metrics.csv`: tabular output for ablation and robustness runs;
2. `research_ablation_summary.csv`: split-aggregated ablation summaries;
3. `research_robustness_summary.csv`: rolling-window comparison statistics;
4. `research_regime_summary.csv`: regime-conditioned action and return summaries;
5. `research_rolling_references.csv`: rolling benchmark metrics for SPY and the factor benchmark;
6. `research_bootstrap_cis.csv`: block-bootstrap confidence intervals for key strategy metrics;
7. `research_bootstrap_significance.csv`: pairwise bootstrap deltas and $p$-value summaries versus benchmark strategies;
8. `research_summary.json`: serialized configuration and best-per-suite summaries; and
9. `research_paper_tables.tex` and `pipeline_research_eval.png`: paper-ready summary tables and figure.

Together with the feature-availability contract embedded in each backtest run, these artifacts make it easier to compare results across alternative timing, cost, and control assumptions without manually reconstructing the experimental context after the fact.

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